Mechanistic Interpretability of Property Representations in Protein Language Models

Introduction

Protein language models (PLMs) learn evolutionary, structural, and biochemical patterns from natural protein sequences [1]. However, it remains unclear whether they encode specific properties directly or indirectly through other learned biological concepts, such as protein family or amino-acid composition.

We investigate direct versus indirect property encoding, a distinction important for understanding whether PLMs learn transferable biochemical knowledge or rely on associations that may fail to generalise to unfamiliar proteins.

Background and Hypotheses

Rives et al. [1] showed that PLMs capture biological information across multiple scales. More recently, InterPLM [2] and InterProt [3] used sparse autoencoders (SAEs) to address the polysemanticity of individual neurons, decomposing activations into sparser, more interpretable features that include family-specific and broadly shared representations. However, identifying property-associated features does not establish whether they represent the property itself or other correlated biological concepts.

Approach and Hypotheses

We investigate four potentially overlapping mechanisms of property encoding:

Hypothesis	**Encoding Mechanism**	**Examples**
H1: Direct	Distinct, property-specific representations	Solubility, Subcellular Localization
H2: Indirect, low-level	Residue-level biochemistry	Amino acid composition, charge, hydrophobicity
H3: Indirect, mid-level	Structural and regional patterns	Aggregation-prone regions, disorder, secondary structure, targeting motifs
H4: Indirect, high-level	Global biological identity	Protein family, homology, phylogeny, fold

We compare solubility, with well-characterised biochemical determinants, and subcellular localisation, influenced by targeting signals and protein identity. Using frozen ESM-2 and pretrained SAEs, we will examine representational overlap, control for correlated biological features, and intervene on internal activations to investigate their functional roles.

Plan

Model: We will study the frozen ESM-2-650M model[4]. Feature-level analysis uses the released InterProt [3] sparse autoencoders for this model.

Dataset: For solubility, we use a stratified subset of approximately 10,000-15,000 proteins from UESolDS (aka PLM Sol) [5], preserving diversity across solubility labels, protein families, and organisms. For localisation we use DeepLoc 2.0 [6], reduced to a secreted-vs-cytoplasmic contrast, whose labels are experimental. We then enrich both datasets with family and clan assignments from Pfam [7], homology clusters from MMseqs2 [8], and source organism from UniProt [9], together with per-level descriptors: composition, charge and hydrophobicity computed from sequence, and aggregation propensity, disorder, secondary-structure content and signal peptides from AGGRESCAN [10], IUPred3 [11], NetSurfP-3.0 [12] and SignalP 6.0 [13], respectively.

Splits: We will retain the published dataset splits: UESolDS provides training, validation, and independent test sets, while DeepLoc 2.0 provides five homology-aware development partitions and an independent Human Protein Atlas test set. We will report test set performance for the final comparison.

Experimental Phases

1. Representation probing (correlational)

We will pass each protein through frozen ESM-2 and pretrained SAEs, extracting raw ESM activations and SAE feature activations across selected layers. We will aggregate residue-level activations into protein-level representations using mean pooling, with region-specific pooling for localised signals such as targeting peptides.

1. Property probing: Train L2-regularised logistic regression classifiers on ESM and SAE activations to distinguish soluble/insoluble or secreted/cytoplasmic proteins. Evaluate predictive performance using held-out AUROC.
2. Biological concept probing: Using the same activations, train separate probes for high-level (family, organism), mid-level (APR burden, disorder, targeting signals), and low-level (composition, charge, hydrophobicity) concepts. Use logistic regression for categorical/binary features and ridge regression for continuous features. Evaluate classification probes using held-out AUROC (or macro-F1 for multiclass targets), and regression probes using held-out R^2.
3. Representation overlap: Compare property and biological concept directions using cosine similarity and subspace projection, accounting for dimensionality. Identify contributing SAE features and check whether associations persist within protein families.

2. Mechanistic intervention (causal)

We will ablate selected SAE features during ESM-2's forward pass and measure changes in property/concept probe outputs and native masked-token predictions.

We will also generate sequence variants using the intervened model and evaluate predicted property changes using independent judges:

- Solubility: CamSol [14] intrinsic solubility score.
- Subcellular localisation: DeepLoc 2.0 extracellular/cytoplasmic predictions, supplemented by SignalP 6.0 for secretion signals.

We will compare these changes against unmodified sequences and matched control interventions to assess whether the identified features have property-specific functional effects.

Timeline and milestones

- Weeks 1-2: Curate datasets, annotate biological features, and extract ESM/SAE activations.
- Check-in 1: Present proposal and datasets.
- Weeks 3-4: Train property and concept probes; analyse representational overlap.
- Check-in 2: Present preliminary probing and overlap results.
- Weeks 5-6: Conduct mechanistic interventions.
- Final presentation: Compare direct versus indirect encoding across both properties.
- **Success criteria:** Quantify property decodability, representational overlap, and the effects of mechanistic interventions.

Risks

Dataset confounding: Control for family, organism, and biochemical features; Computational cost: Use pretrained SAEs and a subset of sequences; Direct versus indirect encoding may not be conclusively distinguishable.

References

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